

Arati Sankaliya

aratisankaliya12@gmail.com | +91 9316586305 | [GitHub](#) | [LinkedIn](#)

Professional Summary

3rd-year B.Tech student in Artificial Intelligence and Data Science (CGPA: 9.57) focused on **Generative AI systems, Agentic workflows, and AI system architecture**. Built an **autonomous Agentic RAG assistant using LangChain and the ReAct framework** while also engineering Linux environments by optimizing Ubuntu boot pipelines and modernizing terminal workflows. Seeking a role where I can **design, build, and deploy production-grade AI systems and autonomous agents**.

Education

A.D. Patel Institute of Technology, CVM University

B.Tech in Artificial Intelligence and Data Science (CGPA: 9.57)

Anand, Gujarat

July 2023 – Present

Technical Skill

- **Programming Languages:** Python, SQL
- **AI / Machine Learning:** PyTorch, HuggingFace, Ollama
- **Generative AI & LLMs:** LangChain, RAG, Agentic Workflows, Prompt Engineering
- **Vector Databases:** ChromaDB, FAISS
- **Backend & Deployment:** FastAPI, Streamlit
- **Data & ML:** Pandas, NumPy, Scikit-learn
- **Developer Tools :** Git, GitHub, Jupyter Notebook
- **Systems & Development Environment:** Linux (Ubuntu), Zsh (Oh-My-Zsh, Starship)

Projects

DigitalTwin – Agentic RAG Personal Assistant

LangChain | OpenAI | ChromaDB | ReAct | Streamlit

[code](#)

- Designed and implemented an **Agentic RAG system** using **LangChain & GPT-4o-mini** that autonomously selects tools (vector retrieval or web search) using the **ReAct reasoning framework**.
- Built a **multi-modal knowledge ingestion pipeline** supporting PDFs, code files, CSVs, and documents, enabling semantic retrieval from personal datasets using **ChromaDB embeddings**.
- Implemented **persistent conversational memory** with FileChatMessageHistory and engineered prompts to reduce LLM hallucinations by **prioritizing local context**.
- Developed an **interactive Streamlit web application** enabling real-time user queries, agent reasoning, and dynamic responses from the RAG pipeline.

MacOS-Inspired Linux Developer Environment

Linux | Systems | Optimization | Automation | Reproducibility

[code](#)

- **Engineered** a macOS-inspired Ubuntu developer environment by optimizing the **boot pipeline (GRUB + systemd)**, reducing startup time from **~30s to ~18s** while preserving system stability.
- **Modernized** terminal workflows by integrating high-performance **CLI tools** (exa, bat, fd, ripgrep, tmux) to replace legacy commands and improve **developer productivity**.
- **Architected** reproducible system configuration using **version-controlled dotfiles, backup/restore scripts, & TimeShift snapshots** to enable safe **experimentation & rapid recovery**.
- **Balanced** aesthetics and reliability through **performance-aware UI customization** (GTK theming, dock behavior, shell enhancements) while intentionally avoiding heavy extensions.

Achievements

Felicitation of Toppers - Dewang Mehta IT Awards 2024

Dewang Mehta IT Awards - Aug 2024

Academic Excellence: First Rank In AI& DS

A.D Patel Institute of Technology - April 2024